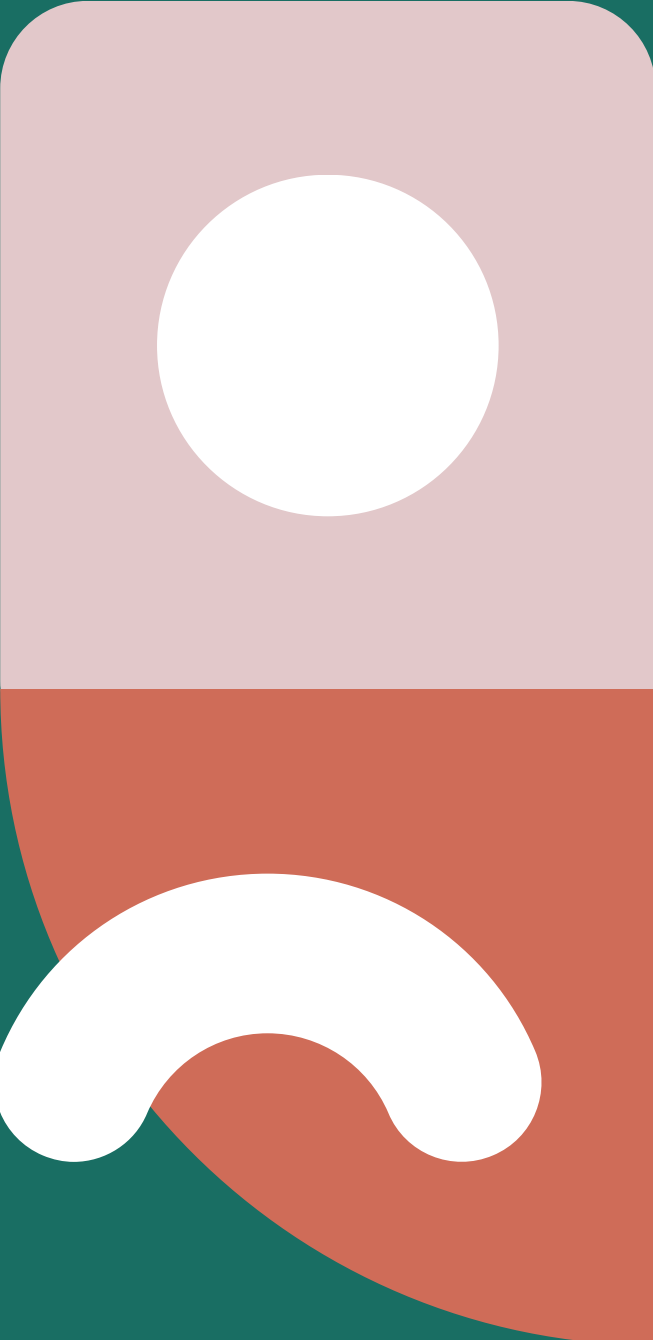
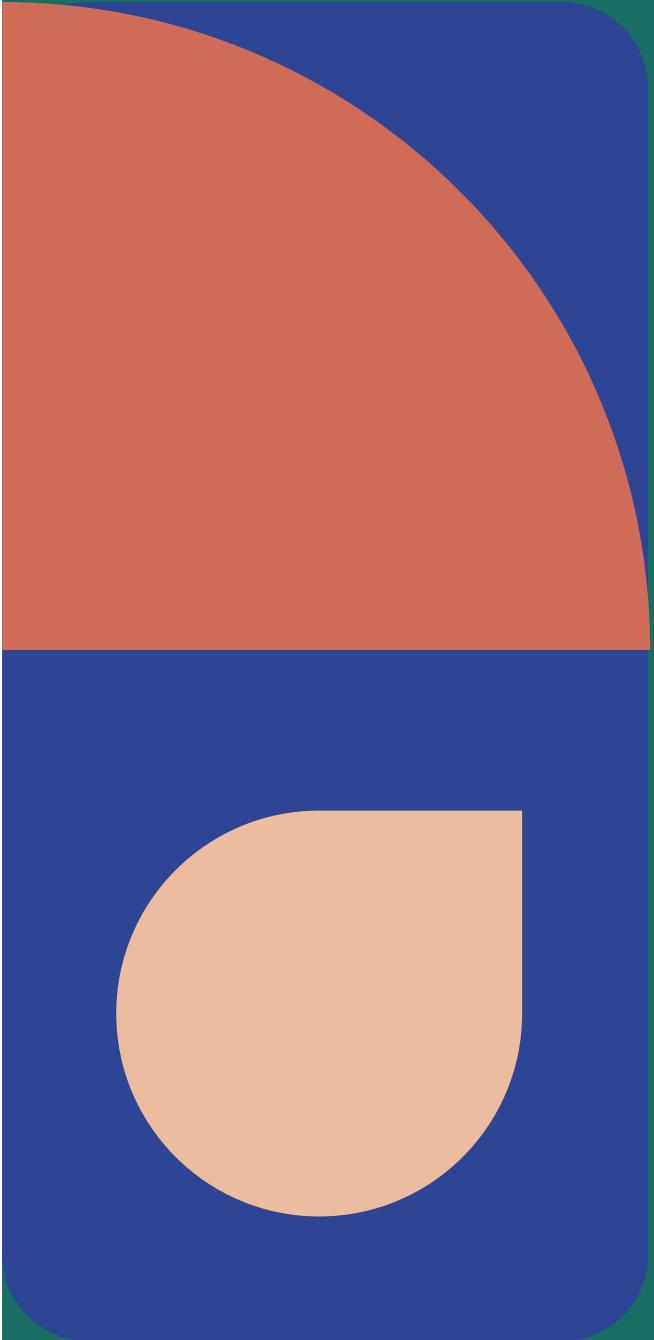




# PREPARE

Advancing Crop Price Prediction and Farmer  
Empowerment



# The Team

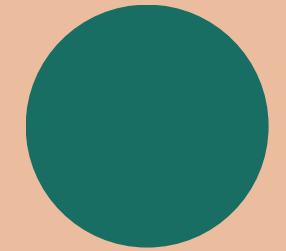
Under Prof.  
Mayank Ratan  
Bhardwaj

**Aryan Gosain**

**Kunal Ranjan**

**Rishit Anand**

**Varun SG**



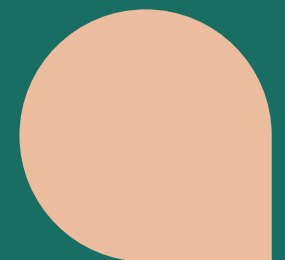
# Problems

## 1: Farmer's Price Blindness

Small and marginal farmers lack timely, accurate price forecasts and struggle to interpret market dynamics.

## 2: Ineffective Government Interventions

Governments and regulators lack reliable multi-day price prediction tools, making it difficult to set fair MSPs or plan timely market interventions.



# The Engine

- Input Data: Historical Price, Geographical Location, Weather
- Data Pre-processing: Outlier Removal, Normalization
- Modeling Core: Define and build upon the model chosen
- Output: Accurate Price Predictions

# The Chassis

- Prediction Service: Refined model is exposed via an API or server.
- Farmer Application: A mobile application for practical deployment.
- End-User Value: Empowers farmers with timely and accurate price forecasts

**Schematic**



## Novelty

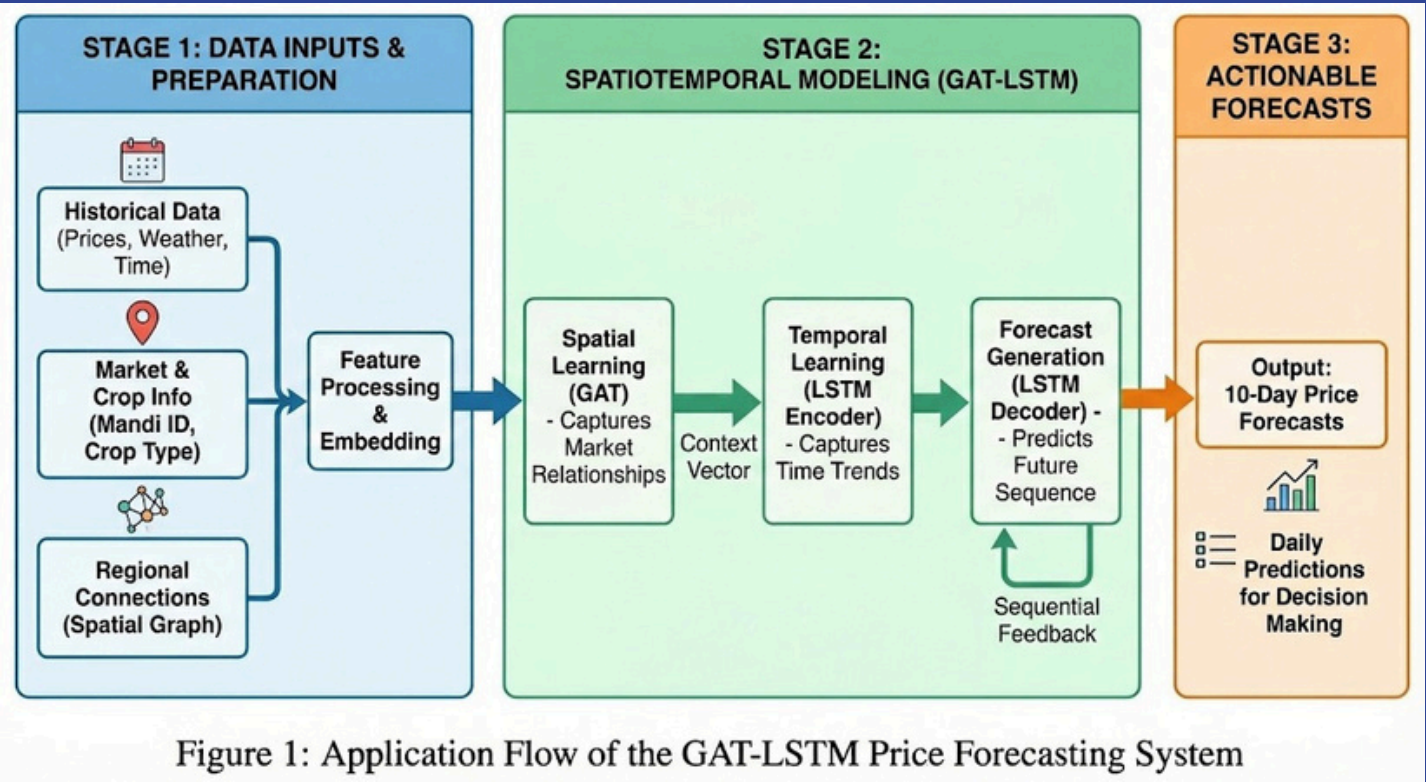
- Capturing the interdependent nature of different markets for food pricing.
- There are **no current consumer facing solutions** that provide price predictions.

## Real World Application

This helps farmers to decide when to harvest crops and where to sell those crops for optimal returns. The format of the application also makes it accessible for farmers and does not have a significant learning curve.

# Model Development and Results

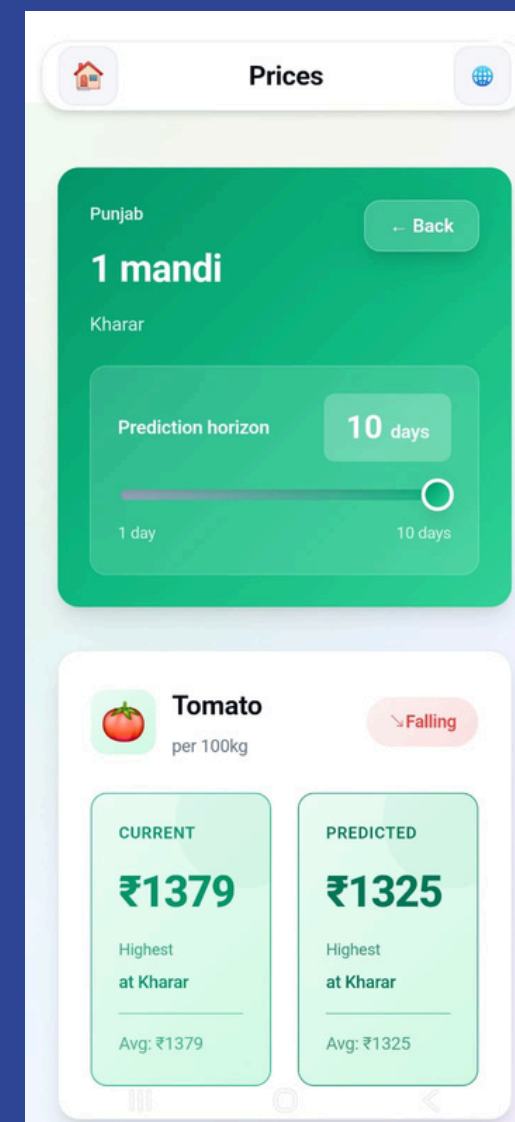
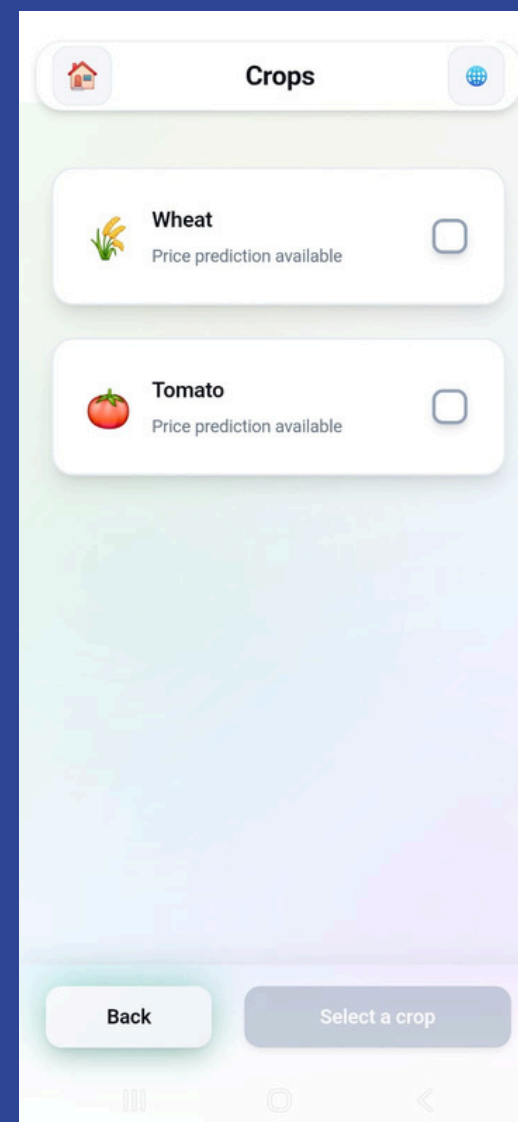
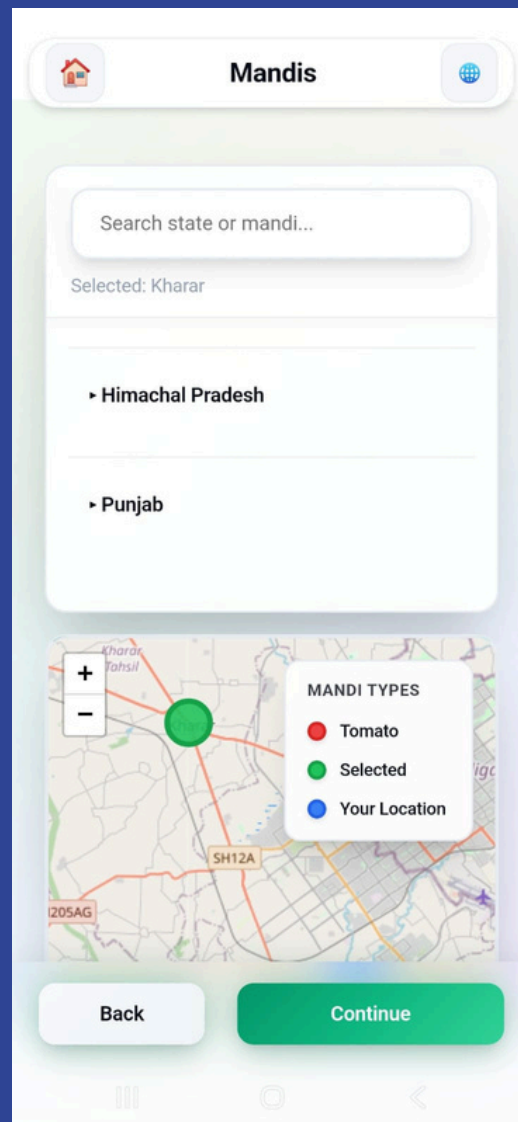
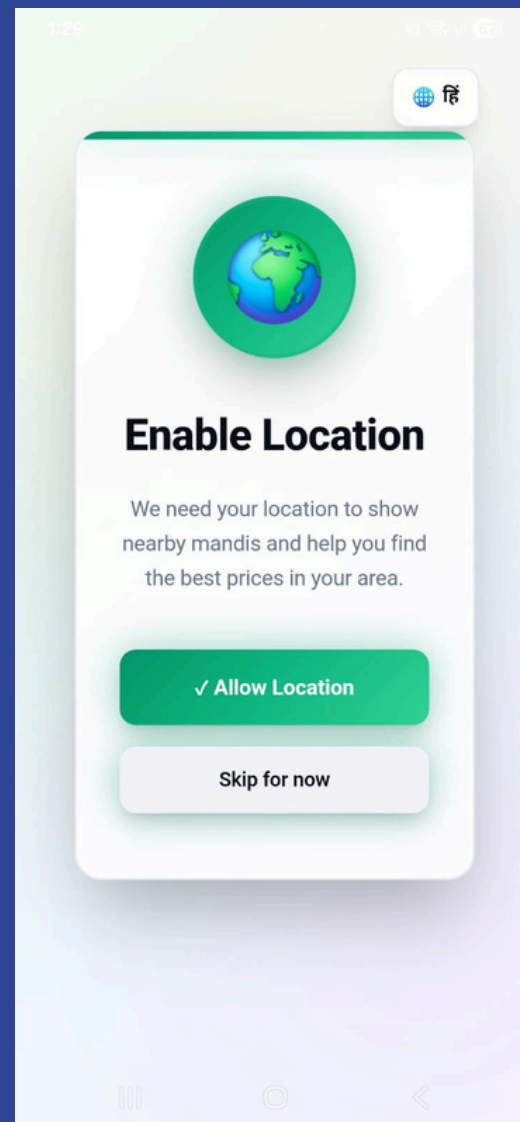
We experimented with and developed models for 10 day price prediction across all APMC mandis in India and settled on a GAT-LSTM model with satisfactory metrics for prediction.



| Metrics |                |               |
|---------|----------------|---------------|
| Day     | RMSE (₹/100kg) | MAE (₹/100kg) |
| 1       | 252.59         | 177.53        |
| 2       | 288.60         | 198.90        |
| 3       | 314.08         | 217.34        |
| 4       | 333.61         | 231.02        |
| 5       | 348.95         | 241.72        |
| 6       | 361.51         | 250.93        |
| 7       | 371.87         | 258.71        |
| 8       | 381.00         | 265.79        |
| 9       | 388.82         | 272.09        |
| 10      | 395.71         | 277.85        |

| Metrics |                |               |
|---------|----------------|---------------|
| Day     | RMSE (₹/100kg) | MAE (₹/100kg) |
| 1       | 146.51         | 76.00         |
| 2       | 158.40         | 75.70         |
| 3       | 167.15         | 79.79         |
| 4       | 174.33         | 81.23         |
| 5       | 179.95         | 82.08         |
| 6       | 183.63         | 82.74         |
| 7       | 185.85         | 83.39         |
| 8       | 187.44         | 84.21         |
| 9       | 188.95         | 85.29         |
| 10      | 190.45         | 86.53         |

# Mobile Interface Development



We developed a mobile interface with features such as location detection, day-based price estimation, and multiple languages and linked our models to the interface using FastAPI.

# Sem 5

Wheat

☐

Mustard

☐

Tomato

☐



Mandis

 हि

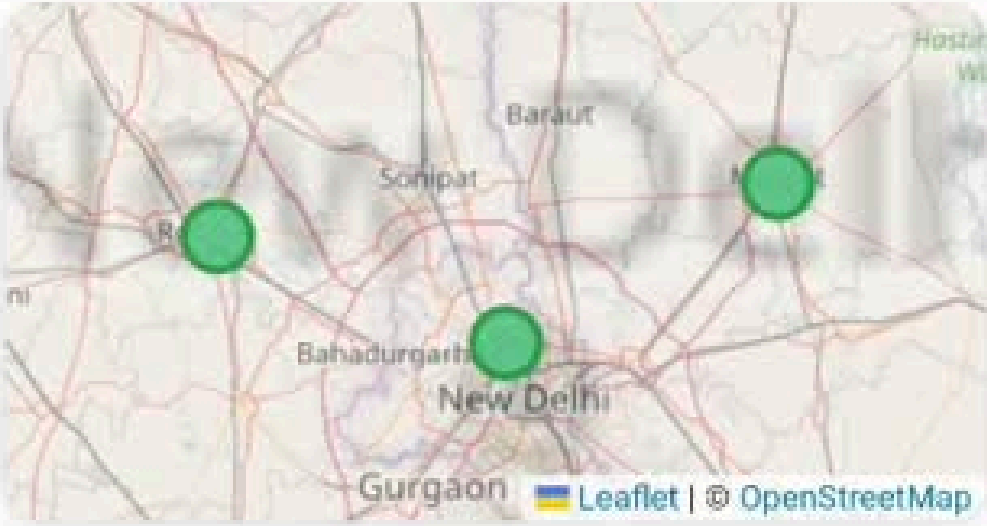
Search state or mandi...

Selected: 3

• Delhi

• Haryana

• Punjab



Leaflet | © OpenStreetMap

Continue

# Data Acquisition

We extracted daily agricultural records from the Agmarknet portal (Jan 2023 – Jan 2026) using custom Python and Playwright scraping scripts. To accurately expose missing data, we expanded the raw extractions into a strict date-by-mandi calendar grid. This ensured every mandi had a dedicated row per day, representing missing arrival quantities and modal prices as explicit null values.



**Government of India**  
**Department of Agriculture & Farmers Welfare**  
**Directorate of Marketing & Inspection**



## Sem 6

Sem 6

Browser window showing the Agmarknet website. The address bar displays `agmarknet.gov.in/all-type-of-report-table`. The page title is "All Type of Report". The header includes the Government of India logo, Department of Agriculture & Farmers Welfare, Directorate of Marketing & Inspection, and the Agmarknet logo. Navigation links include Home, Price Trend Reports, Price & Arrival Reports, Market Profile, Helpdesk, and Contact Us. A "Login" button is visible on the right.

The main content area is titled "All Type of Report (All Grades)". It features a "Back" button on the left and "Main Menu" and "Export" buttons on the right. Below this, there is a table with a "Rows per page" dropdown set to 10. The table has a pagination bar at the bottom showing page numbers 1, 2, 3, 4, 5, and a total of 1568 rows.

A chatbot icon labeled "ChatBot" is located in the bottom right corner of the page.

```

Selecting 'Cereals' for 'Commodity Group'...
Selecting 'Wheat' for 'Commodity'...
Selecting 'Both Arrival Quantity & Weighted Average Modal prices' for 'Arrival and Price'...
Selecting 'Date Wise' for 'Period'...

Scraping data for period: 2023-05-01 to 2023-05-31
Clicking Submit and expecting navigation...
Navigated to results page: https://agmarknet.gov.in/all-type-of-report-table

```

**Top 15 mandis with lowest missing % (most complete):**

| State          | Market                | Days in Range | Present | Missing | % Missing |
|----------------|-----------------------|---------------|---------|---------|-----------|
| Madhya Pradesh | Badamalhera APMC      | 1,065         | 741     | 324     | 30.4%     |
| Uttar Pradesh  | Mugrabaadshahpur APMC | 1,065         | 741     | 324     | 30.4%     |
| West Bengal    | Khatra APMC           | 1,065         | 749     | 316     | 29.7%     |
| Madhya Pradesh | Dhar APMC             | 1,065         | 764     | 301     | 28.3%     |
| Uttar Pradesh  | Tulsipur APMC         | 1,065         | 770     | 295     | 27.7%     |
| Madhya Pradesh | Umariya APMC          | 1,065         | 772     | 293     | 27.5%     |
| Uttar Pradesh  | Jayas APMC            | 1,065         | 772     | 293     | 27.5%     |

We initially targeted five crops, but EDA on the expanded grid revealed severe reporting sparsity. Sugarcane was dropped due to an extreme 98.6% missing data rate. Remaining crops also showed massive gaps; for example, Wheat was missing 75.6% overall and 90.87% on Sundays. We also found hundreds of mandis with uninterrupted missing streaks lasting over a year, prompting us to finalize our scope to Onion, Potato, Tomato, and Wheat.

# Location and Weather Data



We mapped exact latitude and longitude coordinates to every mandi and integrated the Copernicus API to extract historical, daily weather features for each location. This transformed our basic price ledger into a multi-dimensional dataset capable of capturing the real-world variables driving agricultural markets.

## Weather & Location Data

| Column    | Description                                  | Non-null           |
|-----------|----------------------------------------------|--------------------|
| latitude  | Latitude (°)                                 | 1,650,576 (100.0%) |
| longitude | Longitude (°)                                | 1,650,576 (100.0%) |
| t         | Temperature (°C)                             | 1,650,576 (100.0%) |
| tp        | Total Precipitation (m)                      | 1,650,576 (100.0%) |
| ssr       | Surface Solar Radiation (MJ/m <sup>2</sup> ) | 1,650,576 (100.0%) |
| r         | Relative Humidity (%)                        | 1,650,576 (100.0%) |

# Price Imputation

To resolve massive data gaps, we researched and came up with ten imputation strategies (ranging from simple baselines to Matrix Factorization) that we benchmarked at  $\geq 0\%$ ,  $\geq 50\%$ , and  $\geq 75\%$  data presence thresholds. Through this, we identified a day of the week based ratio to be the best performing strategy for mandis with atleast 50% data and so that is what we went with.

Ranking — All Mandis (5,806 mandis total) — Modal\_Price

| Rank | Method            | RMSE (mean) | MAE (mean) | MAPE (%) | Coverage (%) | Time (s) |
|------|-------------------|-------------|------------|----------|--------------|----------|
| 1    | 8_Spline_Pipeline | 396.32      | 105.01     | 5.72     | 99.7         | 105.3    |
| 2    | 2_Rolling_Mean    | 436.22      | 112.78     | 7.75     | 93.14        | 3.36     |
| 3    | 7_DOW_Ratio       | 449.05      | 113.41     | 6.91     | 88.98        | 22.28    |
| 4    | 10_SVD_Matrix     | 476.94      | 157.89     | 9.39     | 88.3         | 459.68   |
| 5    | 9_Random_Forest   | 477.01      | 144.01     | 8.19     | 100.0        | 22.22    |
| 6    | 1_Capped_FFill    | 600.66      | 104.03     | 5.13     | 89.1         | 2.22     |
| 7    | 6_FFill_Decay     | 600.74      | 126.67     | 6.96     | 99.26        | 44.12    |
| 8    | 3_State_Median    | 765.26      | 272.81     | 14.66    | 87.6         | 0.78     |
| 9    | 4_Seasonal        | 921.38      | 410.24     | 28.28    | 87.76        | 0.82     |
| 10   | 5_Global_Median   | 1226.85     | 694.45     | 43.69    | 100.0        | 0.24     |

# Price Imputation



Ranking —  $\geq 50\%$  (2,645 mandis total) — Modal\_Price

| Rank | Method            | RMSE (mean) | MAE (mean) | MAPE (%) | Coverage (%) | Time (s) |
|------|-------------------|-------------|------------|----------|--------------|----------|
| 1    | 7_DOW_Ratio       | 259.58      | 120.58     | 5.78     | 99.7         | 8.2      |
| 2    | 2_Rolling_Mean    | 262.53      | 122.54     | 5.85     | 99.98        | 1.88     |
| 3    | 9_Random_Forest   | 298.68      | 143.01     | 6.88     | 100.0        | 14.35    |
| 4    | 1_Capped_FFill    | 306.27      | 114.15     | 5.44     | 99.05        | 0.95     |
| 5    | 8_Spline_Pipeline | 315.9       | 122.93     | 6.02     | 100.0        | 74.95    |
| 6    | 6_FFill_Decay     | 328.31      | 133.9      | 6.56     | 100.0        | 9.7      |
| 7    | 10_SVD_Matrix     | 398.73      | 169.69     | 8.45     | 100.0        | 83.12    |
| 8    | 4_Seasonal        | 762.14      | 468.0      | 23.77    | 99.25        | 0.3      |
| 9    | 3_State_Median    | 787.07      | 316.48     | 15.54    | 99.25        | 0.3      |
| 10   | 5_Global_Median   | 1272.68     | 793.32     | 42.23    | 100.0        | 0.1      |

Ranking —  $\geq 75\%$  (1,066 mandis total) — Modal\_Price

| Rank | Method            | RMSE (mean) | MAE (mean) | MAPE (%) | Coverage (%) | Time (s) |
|------|-------------------|-------------|------------|----------|--------------|----------|
| 1    | 2_Rolling_Mean    | 202.32      | 96.75      | 4.82     | 100.0        | 0.72     |
| 2    | 7_DOW_Ratio       | 203.38      | 98.7       | 4.97     | 99.93        | 2.17     |
| 3    | 6_FFill_Decay     | 222.32      | 99.0       | 5.22     | 100.0        | 2.25     |
| 4    | 9_Random_Forest   | 226.58      | 111.14     | 5.6      | 100.0        | 4.92     |
| 5    | 1_Capped_FFill    | 227.07      | 85.07      | 4.31     | 99.58        | 0.35     |
| 6    | 8_Spline_Pipeline | 228.27      | 90.97      | 4.76     | 100.0        | 25.12    |
| 7    | 10_SVD_Matrix     | 271.18      | 139.98     | 7.48     | 100.0        | 14.32    |
| 8    | 3_State_Median    | 437.27      | 251.05     | 14.08    | 98.3         | 0.08     |
| 9    | 4_Seasonal        | 617.85      | 383.54     | 20.13    | 99.5         | 0.1      |
| 10   | 5_Global_Median   | 1063.42     | 707.14     | 36.68    | 100.0        | 0.02     |

# Baseline Model



Used a baseline LightGBM regressor to predict prices for one day, we will extend this model to multiple days (15 days)

**Time-Aware Feature Engineering:** Significant performance gains were achieved by augmenting location and weather data with historical price lags (1 to 14 days) and moving averages (7 to 30 days) per market.

**Performance Dynamics:** The model predicts stable commodities exceptionally well (Potato  $R^2 = 0.95$ , Onion  $R^2 = 0.89$ ) but struggles more with the higher volatility of highly perishable crops (Tomato  $R^2 = 0.74$ ).

| Commodity | MAE    | RMSE   | $R^2$  |
|-----------|--------|--------|--------|
| Tomato    | 463.08 | 683.10 | 0.7426 |
| Potato    | 107.32 | 311.72 | 0.9552 |
| Onion     | 174.71 | 517.70 | 0.8917 |
| Wheat     | 45.95  | 80.71  | 0.8404 |

**Game Theory**

**Improve 10-day models  
and introduce new  
seasonal/cash crops.**

**Cluster Mandis by  
Production Volume and  
Develop Models**

**Explainable AI**

# Individual Contribution

## Aryan Gosain

- Engineered the strict Date × Mandi calendar grid to accurately expose explicit null values.
- Restructured the data loaders and pre-processing pipelines to integrate the newly imputed, multi-dimensional dataset into the GAT-LSTM architecture for 15-day forecasting.

## Kunal Ranjan

- Mapped exact latitude and longitude coordinates for all targeted APMC mandis.
- Integrated the Copernicus API to extract historical, daily meteorological features (temperature, precipitation, solar radiation) for the 3-year period.

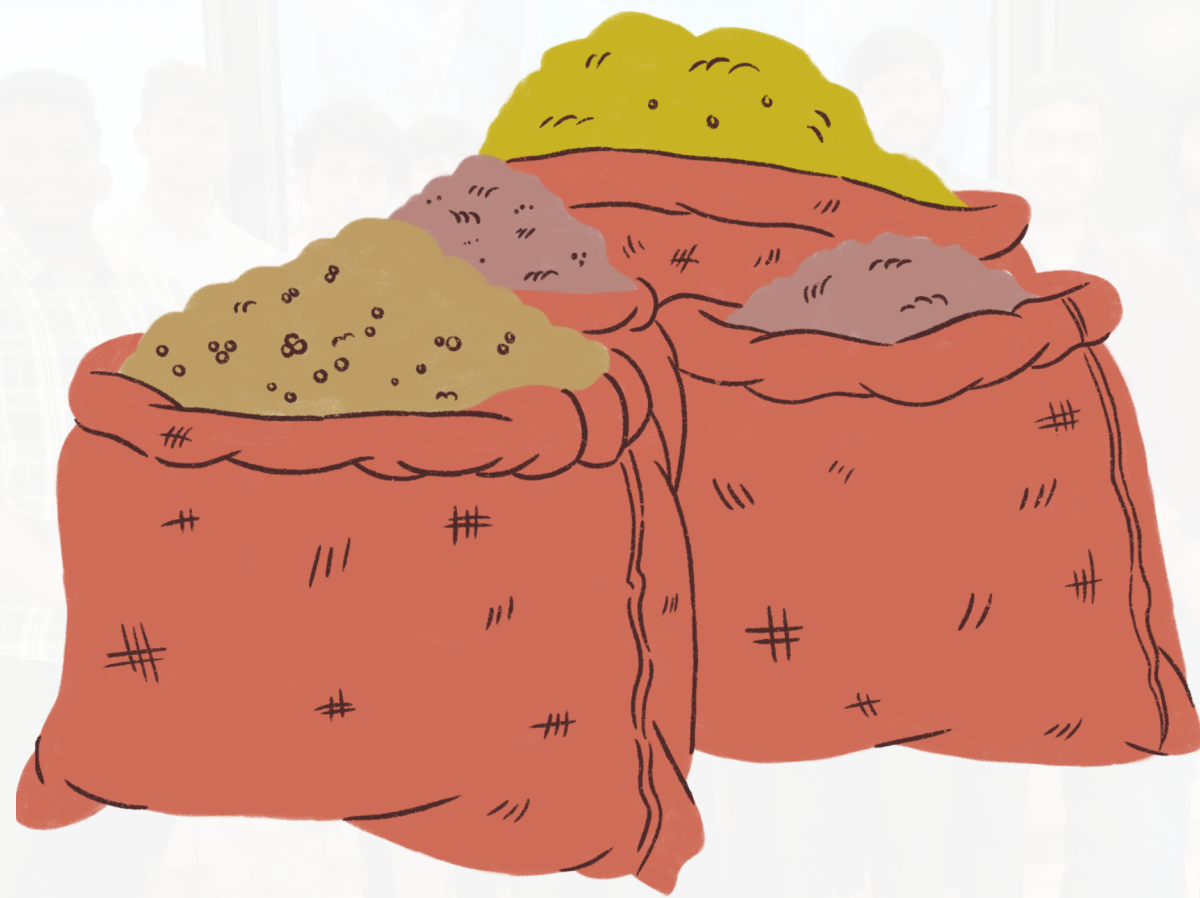
## Rishit Anand

- Conducted rigorous EDA on the expanded dataset, providing the data-backed justification to drop Sugarcane (98.6% missing).
- Developed the pipeline to benchmark 10 imputation strategies at  $\geq 0\%$ , 50%, and 75% data presence thresholds, finalizing the Day-of-Week Ratio strategy.

## Varun SG

- Architected custom Python and Playwright scraping scripts to extract daily agricultural records.
- Managed the extraction of 3 years of Agmarknet data (Jan 2023 - Jan 2026), navigating complex portal rate limits.

thank



you